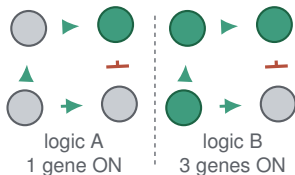


Logic over wiring: the model reads local Boolean logic, not just the wiring.

TRAIN DATA

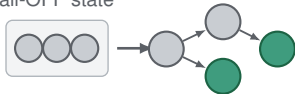
Synthetic Boolean nets ($n \leq 16$)

Same wiring, different logic
→ **different reachability**



Exact oracle

all-OFF state

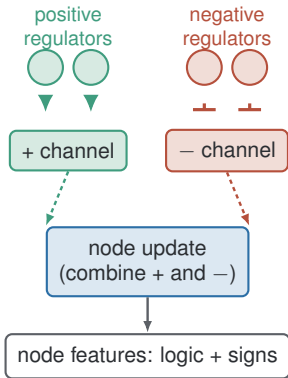


→ per-node reachability labels

→ activation —| inhibition

MODEL

Sign-aware, size-agnostic GNN

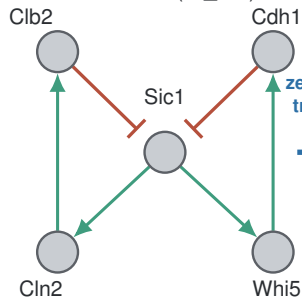


trained on synthetic logic only

● reachable / ON

TRANSFER

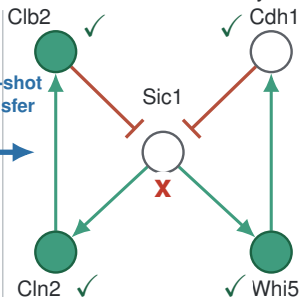
Real GRNs ($n \leq 70$)



Exact check
symbolic BDD, validation only

OUTPUT

Predicted activatability



Coloured by prediction:

- activatable
- not activatable
- ✓ matches exact oracle
- X differs from oracle

AUROC 0.91 [0.88–0.93]